Agriculture

The big idea

India is an agriculturally important country: about **two-thirds** of its population is engaged in agricultural activities. Agriculture produces most of the food we eat and raw materials for industries (cotton, jute, sugarcane). This chapter looks at **types of farming**, **cropping seasons**, **major crops** with the conditions they need and where they grow, and the **reforms** that aim to make farming productive and fair.

Types of farming

1. PRIMITIVE SUBSISTENCE FARMING

Practised on small patches of land with primitive tools like the hoe, dao and digging sticks, and family/community labour. Depends on monsoon, natural fertility of soil and suitability of other environmental conditions. It is **'slash and burn' agriculture**: farmers clear a patch of land, produce cereals and other food crops, and when soil fertility decreases, shift to a new patch. Nature replenishes the fertility of the old patch. Productivity is low because no fertilisers or modern inputs are used.

Names of shifting cultivation:

In India	Outside India
Jhumming in Assam, Meghalaya, Mizoram, Nagaland (north-east)	**Milpa** in Mexico and Central America
Bewar or **Dahiya** in Madhya Pradesh	**Conuco** in Venezuela

In India	Outside India
Podu or **Penda** in Andhra Pradesh	**Roca** in Brazil
Pama Dabi, **Koman** or **Bringa** in Odisha	**Masole** in Central Africa
Kumari in the Western Ghats	**Ladang** in Indonesia
Valre or **Waltre** in south-eastern Rajasthan	**Ray** in Vietnam
Khil in the Himalayan belt; **Kuruwa** in Jharkhand	

2. INTENSIVE SUBSISTENCE FARMING

Practised in areas of **high population pressure on land**. It is **labour-intensive**; high doses of biochemical inputs and irrigation are used for higher production. Due to the 'right of inheritance', land is divided among successive generations, so landholdings become **uneconomical** and small. Farmers continue to take maximum output from limited land because they have no alternative source of livelihood.

3. COMMERCIAL FARMING

Uses **higher doses of modern inputs** like high yielding variety (HYV) seeds, chemical fertilisers, insecticides and pesticides to obtain higher productivity. The degree of commercialisation varies by region: **rice is a commercial crop in Punjab and Haryana**, but a subsistence crop in Odisha. **Plantation** is a type of commercial farming in which a **single crop** is grown on a large area. It has an interface of agriculture and industry, uses capital-intensive inputs and migrant labourers, and all produce is used as raw material in industries. Examples: **tea, coffee, rubber, sugarcane, banana**. Tea in **Assam and North Bengal**, coffee in **Karnataka**. Development of plantations depends on a well-developed network of transport and communication connecting plantations, processing industries and markets.

Cropping seasons

Season	Sown	Harvested	Main crops
Rabi	Winter, **October to December**	Summer, **April to June**	**Wheat, barley, peas, gram, mustard**
Kharif	Onset of **monsoon** (June–July)	**September–October**	**Paddy, maize, jowar, bajra, tur (arhar), moong, urad, cotton, jute, groundnut, soybean**
Zaid	Short season **between rabi and kharif** (summer months)		**Watermelon, muskmelon, cucumber, vegetables, fodder crops**

- **Rabi** crops are grown mainly in the north and north-western states: Punjab, Haryana, Himachal Pradesh, Jammu and Kashmir, Uttarakhand and Uttar Pradesh. **Western temperate cyclones** (western disturbances) bring winter precipitation that helps the success of these crops; the Green Revolution in Punjab, Haryana, western UP and parts of Rajasthan also helped.
- **Kharif**: important rice-growing regions include Assam, West Bengal, coastal Odisha, Andhra Pradesh, Telangana, Tamil Nadu, Kerala and Maharashtra (Konkan), along with UP and Bihar; recently paddy has also become important in Punjab and Haryana. In **Assam, West Bengal and Odisha**, three crops of paddy are grown in a year: **Aus, Aman and Boro**.
- **Sugarcane** takes almost a year to grow.

Major crops

FOOD CROPS

Rice - staple food of the majority of people. India is the **second largest producer of rice** in the world after **China**. Kharif crop; needs **high**

temperature (above 25°C), **high humidity** and **annual rainfall above 100 cm**; grown in areas with less rainfall with the help of irrigation. Grown in the plains of north and north-east India, coastal areas and deltaic regions. Canal irrigation and tube-wells made it possible in Punjab, Haryana, western UP and parts of Rajasthan.

Wheat - second most important cereal; main food crop in the north and north-west. Rabi crop; needs a **cool growing season and bright sunshine at the time of ripening**; **50 to 75 cm** of annual rainfall evenly distributed over the growing season. Two important wheat-growing zones: the **Ganga-Satluj plains** in the north-west and the **black soil region of the Deccan**. Major producers: Punjab, Haryana, UP, MP, Bihar and Rajasthan.

Millets - **jowar, bajra and ragi**: coarse grains with very high nutritional value. **Ragi** is very rich in **iron, calcium**, other micro-nutrients and roughage.

- **Jowar**: third most important food crop with respect to area and production; **rain-fed**, grown in moist areas that hardly need irrigation. **Maharashtra** is the largest producer, followed by Karnataka, Andhra Pradesh and Madhya Pradesh.
- **Bajra**: grows well on **sandy soils and shallow black soil**. **Rajasthan** is the largest producer, followed by UP, Maharashtra, Gujarat and Haryana.
- **Ragi**: a crop of **dry regions**; grows well on red, black, sandy, loamy and shallow black soils. **Karnataka** is the largest producer, followed by Tamil Nadu; also HP, Uttarakhand, Sikkim, Jharkhand and Arunachal Pradesh.

  Maize - used both as food and fodder. Kharif crop requiring **21°C to 27°C** and old alluvial soil. In some states like **Bihar**, maize is grown in the **rabi** season too. Major producers: Karnataka, Madhya Pradesh, UP, Bihar, Andhra Pradesh and Telangana. Use of modern inputs (HYV seeds, fertilisers, irrigation) increased production.

  Pulses - India is the **largest producer as well as consumer of pulses** in the world. They are a major source of **protein** in a vegetarian diet. Major pulses: tur (arhar), urad, moong, masur, peas and gram. Pulses need less moisture and survive even in dry conditions. Being **leguminous** crops, all except arhar help **restore soil fertility by fixing nitrogen** from the air, so

they are mostly grown in rotation with other crops. Major producers: Madhya Pradesh, Uttar Pradesh, Rajasthan, Maharashtra and Karnataka.

FOOD CROPS OTHER THAN GRAINS

Sugarcane - a tropical as well as subtropical crop. Grows well in hot and humid climate with **21°C to 27°C** and **75–100 cm** annual rainfall; irrigation needed in low-rainfall regions. Grown on a variety of soils; needs manual labour from sowing to harvesting. India is the **second largest producer of sugarcane after Brazil**. It is the main source of sugar, gur (jaggery), khandsari and molasses. Major producers: **Uttar Pradesh**, Maharashtra, Karnataka, Tamil Nadu, Andhra Pradesh, Telangana, Bihar, Punjab and Haryana.

Oilseeds - India is among the largest producers of oilseeds in the world; different oilseeds cover approximately 12 per cent of the total cropped area. Main oilseeds: **groundnut, mustard, coconut, sesamum (til), soybean, castor seeds, cotton seeds, linseed and sunflower**. **Groundnut** is a kharif crop and accounts for about half of the major oilseeds produced; **Gujarat** was the largest producer of groundnut (2019–20 figures in NCERT), followed by Rajasthan and Tamil Nadu. **Linseed and mustard** are rabi crops. **Sesamum** is a kharif crop in the north and a rabi crop in the south. **Castor seed** is grown both as rabi and kharif.

Tea - a beverage crop introduced in India initially by the **British**. Today, most tea plantations are owned by Indians. Grows well in **tropical and subtropical climates** endowed with **deep and fertile well-drained soil, rich in humus and organic matter**. Tea bushes require a **warm and moist frost-free climate** all through the year; frequent showers evenly distributed over the year ensure continuous growth of tender leaves. Tea is **labour-intensive** and requires abundant, cheap and skilled labour. It is processed within the tea garden to retain freshness. Major producers: **Assam**, the hills of **Darjeeling and Jalpaiguri** districts (West Bengal), Tamil Nadu and Kerala; also HP, Uttarakhand, Meghalaya, Andhra Pradesh and Tripura.

Coffee - Indian coffee is known for its good quality. The **Arabica** variety initially brought from **Yemen** is produced in the country; it is in great

demand worldwide. Its cultivation was introduced on the **Baba Budan Hills**, and even today it is confined to the **Nilgiris** in **Karnataka, Kerala and Tamil Nadu**.

Horticulture crops - India is the **second largest producer of fruits and vegetables** in the world (after China). Mangoes of Maharashtra, Andhra Pradesh, Telangana, UP and West Bengal; oranges of Nagpur and Cherrapunjee (Meghalaya); bananas of Kerala, Mizoram, Maharashtra and Tamil Nadu; litchi and guava of UP and Bihar; pineapples of Meghalaya; grapes of Andhra Pradesh, Telangana and Maharashtra; apples, pears, apricots and walnuts of Jammu and Kashmir and Himachal Pradesh.

NON-FOOD CROPS

Rubber - an **equatorial crop**, but under special conditions also grown in tropical and subtropical areas. Requires **moist and humid climate** with rainfall of **more than 200 cm** and temperature **above 25°C**. An important industrial raw material. Mainly grown in **Kerala**, Tamil Nadu, Karnataka, **Andaman and Nicobar Islands** and the **Garo hills of Meghalaya**.

Fibre crops - **cotton, jute, hemp and natural silk**. The first three are derived from crops grown in the soil; silk is obtained from cocoons of silkworms fed on green leaves of mulberry. Rearing of silkworms for silk fibre is called **sericulture**.

Cotton - India is believed to be the original home of the cotton plant. India is the **second largest producer of cotton** in the world (after China). Grows well in **drier parts of the black cotton soil of the Deccan plateau**. Requires **high temperature, light rainfall or irrigation, 210 frost-free days and bright sunshine**. It is a **kharif** crop and requires **6 to 8 months** to mature. Major producers: **Maharashtra, Gujarat**, Madhya Pradesh, Karnataka, Andhra Pradesh, Telangana, Tamil Nadu, Punjab, Haryana and Uttar Pradesh.

Jute - known as the **golden fibre**. Grows well on **well-drained fertile soils in flood plains where soils are renewed every year**. High temperature is required during growth. Major producers: **West Bengal**, Bihar, Assam, Odisha and Meghalaya. Used for gunny bags, mats, ropes, yarn, carpets

and other artefacts. Due to its high cost, it is losing market to synthetic fibres and packing materials, especially nylon.

Technological and institutional reforms

Sustained uses of land without compatible techno-institutional changes have hindered agricultural development. Despite development of irrigation, farmers in many parts still depend on monsoon and natural fertility.

- **Institutional reforms** after independence: collectivisation, **consolidation of holdings**, cooperation and **abolition of zamindari** were given priority in the First Five Year Plan.
- **Land reforms** were the main focus of the First Five Year Plan; the right of inheritance had led to fragmentation of holdings.
- In the 1960s and 1970s, the Government of India introduced agricultural reforms: the **Green Revolution** (based on HYV seeds, chemical fertilisers and irrigation) and the **White Revolution (Operation Flood)** to improve milk production.
- **Comprehensive land development programme** included both institutional and technical reforms: provision for **crop insurance** against drought, flood, cyclone, fire and disease; establishment of **Grameen banks, cooperative societies and banks** for loan facilities at lower interest rates.
- **Kisan Credit Card (KCC)** and **Personal Accident Insurance Scheme (PAIS)** were introduced.
- Special weather bulletins and agricultural programmes on radio and television.
- The government announces **Minimum Support Price (MSP)**, remunerative and procurement prices for important crops to check exploitation of farmers by speculators and middlemen.

BHOODAN-GRAMDAN

Vinoba Bhave, a spiritual heir of Gandhi, undertook padyatra to spread Gandhi's message. At **Pochampalli** (now in Telangana), some landless villagers demanded land. Vinoba Bhave promised to try, and **Shri Ram Chandra Reddy** offered **80 acres** of land to 80 landless villagers. This act was known as **'Bhoodan'**. Some zamindars, owners of many villages, offered to distribute entire villages among the landless; this was known as **'Gramdan'**. This initiative is also known as the **"Blood-less Revolution"**.

Contribution of agriculture and food security. Agriculture is the backbone of the Indian economy, but its share in GDP has declined. The government has set up the **Food Corporation of India (FCI)**, which procures food grains from farmers at MSP, and the **Public Distribution System (PDS)** distributes them at subsidised prices through ration shops.

SECOND-LARGEST PRODUCER MIX-UPS

India is **second** (after China) in rice, cotton, fruits and vegetables, and tea (after China); **second** in sugarcane after **Brazil**; but the **largest** producer **and** consumer of **pulses**.

BOARD FAVOURITES AND MAP WORK

"Compare intensive subsistence and commercial farming", "Distinguish rabi and kharif", "Geographical conditions for rice/wheat/tea/cotton/sugarcane" (3 marks each), "Technological and institutional reforms" (5 marks), "What was Bhoodan-Gramdan?". **Map:** identify major areas of **rice and wheat**, the largest/major producer states of **sugarcane, tea, coffee, rubber, cotton and jute**.

Quick recap

- Farming types: primitive subsistence (slash and burn, jhumming), intensive subsistence, commercial (incl. plantation).

- Rabi (Oct–Dec sow; wheat, mustard), kharif (monsoon; rice, cotton, jute), zaid (summer; melons, fodder).
- Rice: > 25°C, > 100 cm rain; wheat: 50–75 cm, cool growing season, bright sunshine at ripening.
- Cotton on black soil, 210 frost-free days; jute (golden fibre) in flood plains; rubber > 200 cm rain.
- Reforms: land reforms, Green and White Revolutions, KCC, PAIS, MSP, crop insurance, Bhoodan-Gramdan.

Practice questions

1. 'Slash and burn' agriculture is called 'Podu' or 'Penda' in
   (a) Madhya Pradesh (b) Andhra Pradesh (c) Odisha (d) Jharkhand
2. Which of these is a rabi crop?
   (a) Paddy (b) Cotton (c) Jowar (d) Mustard
3. Aus, Aman and Boro are three crops of
   (a) wheat (b) cotton (c) paddy (d) jute
4. India is the second largest producer of sugarcane after
   (a) China (b) USA (c) Cuba (d) Brazil
5. Which millet is very rich in iron, calcium and roughage and is mainly grown in Karnataka?
   (a) Jowar (b) Bajra (c) Ragi (d) Maize
6. Cotton requires how many frost-free days?
   (a) 210 (b) 100 (c) 150 (d) 300
7. The Bhoodan movement began when Ram Chandra Reddy offered land to landless villagers at
   (a) Champaran (b) Sabarmati (c) Pochampalli (d) Bardoli
8. Describe the geographical conditions required for the growth of tea.
9. Describe the technological and institutional reforms introduced in Indian agriculture after independence.

Answers: 1. b 2. d 3. c 4. d 5. c 6. a 7. c 8. see notes 9. see notes